

Quantum Q&A — Question & Answer Format

Below are the questions (Q1, Q2, ...) followed immediately by clear, short answers for easy study. Use this for quick revision or copy into Canva.

Right-column questions (Q1–Q12)

Q1. China's famous quantum satellite is named — ?

A1. Micius (also spelled *Mozi*).

Q2. Measurement in quantum mechanics results in — ?

A2. Wavefunction collapse: the quantum state randomly collapses to an eigenstate of the measured observable.

Q3. The wave nature of matter was proposed by — ?

A3. Louis de Broglie.

Q4. Quantum teleportation refers to — ?

A4. The **transfer of an unknown quantum state** between particles using entanglement plus classical communication.

Q5. Superposition is used in quantum computing to — ?

A5. Let qubits be in **multiple states simultaneously**, enabling quantum parallelism and speedups for some tasks.

Q6. In quantum theory, measurement causes — ?

A6. Collapse or disturbance of the quantum state and yields a definite probabilistic outcome.

Q7. Which country demonstrated the first space-based quantum key distribution? — ?

A7. China (using the Micius satellite).

Q8. Which is a major challenge in quantum computing? — ?

A8. Decoherence and error correction: keeping qubits coherent and correcting errors is the biggest practical hurdle.

Q9. Quantum key distribution (QKD) is mainly used for — ?

A9. Secure key exchange for cryptographic communication (information-theoretic security in some protocols).

Q10. Quantum communication uses — ?

A10. Photons and entanglement (single photons or entangled photon pairs) to transmit quantum information.

Q11. Which pair of quantities is subject to the uncertainty principle? — ?

A11. Position and momentum ($\Delta x \cdot \Delta p \geq \hbar/2$). Energy–time is an analogous relation.

Q12. Ground state of an atom is — ?

A12. The **lowest-energy state** of the atom (e.g., hydrogen's $n=1$ level).

Additional short Q&A (left column / extras)

Q13. What does superposition mean?

A13. A particle can be in **more than one quantum state at once** until measured.

Q14. What does the EU Quantum Flagship focus on?

A14. A large EU program to develop quantum technologies (computing, communication, sensing); funded at about **€1 billion** over its duration.

Q15. When is quantum mechanics essential?

A15. At **atomic and subatomic scales** where classical physics fails (discrete energy levels, tunnelling, etc.).

Q16. Who introduced quantization first (and why)?

A16. Max Planck introduced quantized energy (to solve blackbody radiation), leading to discrete energy levels.

Q17. Limitation(s) of classical mechanics?

A17. It **cannot explain atomic stability or quantized spectra** — quantum theory was needed.

Q18. Aim of national quantum missions (e.g., India)?

A18. Build capabilities in **quantum computing, secure quantum communication networks, sensing**, and develop qubit technology.

Q19. What is a qubit?

A19. A **quantum bit** able to be $|0\rangle$, $|1\rangle$, or any superposition $\alpha|0\rangle + \beta|1\rangle$; basic unit of quantum information.

Q20. Which system is best for quantum communication?

A20. Photons (optical photons) — ideal for long-distance quantum communication and QKD.

Q21. Planck's constant relates what?

A21. Energy and frequency: $E = h \cdot \nu$ (h is Planck's constant).

Q22. Who proposed the uncertainty principle?

A22. Werner Heisenberg.

Q23. What is wave-particle duality?

A23. Particles (electrons, photons) exhibit **wave-like** behaviour (interference) and **particle-like** behaviour (localized detection) depending on the experiment.

If you want a version arranged as two columns (questions left, answers right) or a printable one-page PDF or PNG for direct upload to Canva, tell me which format and I'll export it for you.